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Investigation of Kiaman Magnetic Divison in Colorado Redbeds

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Summary

Palaeomagnetic examination of Middle and Upper Pennsylvanian, Permian and Triassic redbeds from the western and eastern slopes of the Colorado Front Range suggests that a late Pennsylvanian–Permian reversed interval equivalent to the Kiaman magnetic interval has been located on the North American continent. Collections made from a stratigraphic interval of 2000 ft on the eastern slope and 4000 ft on the western slope showed reversed directions only. Available stratigraphic data indicate that the zone extends from the beginning of Late Pennsylvanian to Late Permian or into Early Triassic. This time span is in good agreement with the age assignment, based on radioisotope dating and stratigraphic position, of the Australian igneous rocks on which the magnetic interval was defined. Thus it appears that the redbeds of this study acquired their magnetic direction at the time of deposition or very soon thereafter.

Palaeomagnetic poles calculated on the basis of these data are: Middle Pennsylvanian, 39° N, 105° E; Upper Pennsylvanian and Permian, 33° N, 126° E, for the western slope and 48° N, 119° E, for the eastern slope; Lower Triassic, 56° N, 100° E.

Introduction

Graham (1955) noted the predominance of reversed directions in Permian redbeds of the western United States. A similar observation was made on the basis of a compilation of magnetic directions by Cox & Doell (1960). Following an examination of Permian lavas in Australia in which only reversed magnetic directions were observed, Irving & Parry (1963) defined the Kiaman magnetic interval, and on the basis of published data from other continents, concluded that the interval extended from the beginning of Late Pennsylvanian time until approximately the close of the Palaeozoic Era. Although the information available from Europe and North America appears to support the presence of a prolonged field reversal, the division has previously been spanned only in the Australian section. In Australia the units studied represent only a small portion of the time interval involved (Irving 1966). The present study was undertaken with the express purpose of spanning the stratigraphic zone which corresponds to the Kiaman interval. We believe that the rock units represent as complete a depositional sequence as it will be possible to find within

a single depositional basin. It was hoped that the data would provide more precise information about the limits of the division.

Samples were collected from two areas in Colorado, one on the eastern slope and one on the western slope of the Front Range of the Rocky Mountains, where there is a thick sequence of redbeds. On the eastern slope, a stratigraphic thickness of approximately 2000 ft was sampled at five sections between Boulder, Colorado and 20 miles north of Boulder. On the western slope, 6000 ft were sampled at three sections located between 35 and 55 miles east of Glenwood Springs (Table 2). Formations sampled in the Boulder area were the Fountain, Lyons and Lykins formations; those of the western slope were the Minturn and Maroon formations. The stratigraphic interval that was sampled ranges in age from Middle Pennsylvanian through Early Triassic (Fig. 1).

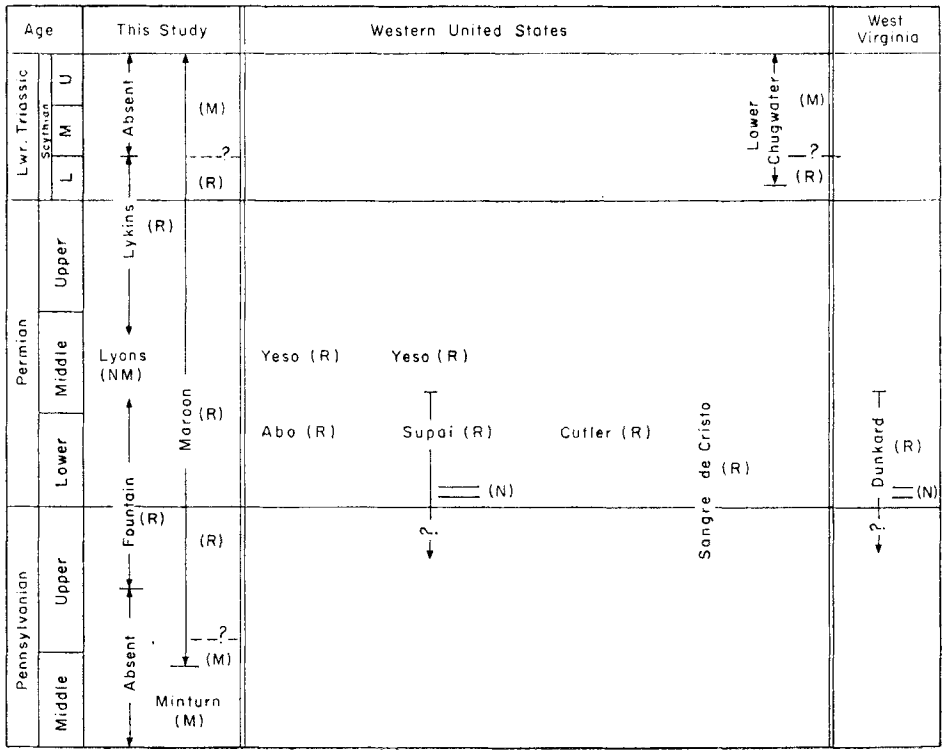


FIG. 1. Palaeomagnetic data from Late Palaeozoic formations, United States.

Field and laboratory procedures

Samples were collected at stratigraphic intervals of 100 ft where possible, but the absence of suitable outcrops made it impossible to obtain this control at all localities. A spinner magnetometer was used to measure the magnetic directions and intensities. In many cases it was necessary to spin the samples in field-free space to eliminate the effects of susceptibility anisotropy which could occasionally be detected. In addition it was necessary to spin most of the samples twice about each axis in order to get meaningful data. This is probably the result of non-uniform distribution of magnetic material in the sample. A similar effect has been reported by Collinson (1965) using astatic magnetometers. An a.c. demagnetization coil capable of gen-

erating a field strength of 500 Oe was availale to test for stability of the magnetism in the specimens.

Usually one sample was taken at each collection site, although at a few sites, two samples were taken. The samples were cut into four to eight specimens suitable for use in the spinner magnetometer. In obtaining the original palaeomagnetic data an average of five specimens from each collection site was measured for palaeomagnetic data. Subsequent sample collections were made, both from the original sections, and from additional sections in order to test the pattern that the original data provided. In the case of these later collections, only two or three of the specimens cut from the samples were measured. When a.c. demagnetization was necessary to clean the specimens, only two specimens were subjected to the cleaning, and only these specimens were used in the directional analyses. A mean direction for each collection site was obtained, each specimen being given unit weight. For each section, a mean magnetic direction for the appropriate zone (e.g. Pre-Kiaman, Kiaman and Post-Kiaman) was calculated giving each site unit weight. A total of 310 specimens from 88 sites were used in the final calculations.

Stability tests

As a test of the stability of the magnetization in the rocks, the Fountain formation was sampled in the vicinity of a rhyolitic intrusion of early Tertiary age. The intrusive is believed to have been emplaced after the rocks were folded to their present position. The magnetic direction of the intrusive confirms this interpretation, as it has a direction close to that of the present field (1 on Fig. 2(a)). Only a few inches of the arkose which is immediately adjacent to the intrusive show loss of colour as a result of baking by the intrusive. However, samples taken from the visible baked zone and one foot away from the baked zone have magnetic directions which reflect the magnetic direction of the intrusive (2 and 4 on Fig. 2(a)). The samples at site 2 had scattered directions until demagnetized in fields of 200 Oe (Fig. 2(b)). Samples at site 3 had high intensities, large scatter (Table 2) and did not respond to demagnetization in fields of 300 Oe. The magnetic direction of the intrusive differs from the present-day field by approximately 13° (see Table 2). This direction was imposed on the adjacent sedimentary rocks about 60 million years ago, and is in good agreement with other magnetic directions for rocks of this age. Basal Fountain material collected 90 and 155 ft below the intrusive has a mean azimuth and inclination of 148° and $-1{\cdot}1°$ respectively, typical of the Permian directions (Fig. 5(e)).

Comparison of the average strike and dip of the various sections sampled (Table 1) shows that the present orientation of the rock units varies greatly from section to section. The present attitudes of the formations date from the Laramide orogeny

Table 1

Average strike and dip for each section

Section	Strike	Dip
Dowe Pass	North	15° E
Pine Brook Hills	N 10° W	73° NE
Lee Hill Road	North	45° E
Six Mile	N 5° W	27° NE
Flagstaff	N 5° W	52° NE
Red Sandstone Creek	N 66° E	27° NW
Squaw Creek	N 86° W	43° NE
Horn Ranch	N 48° W	11° NE

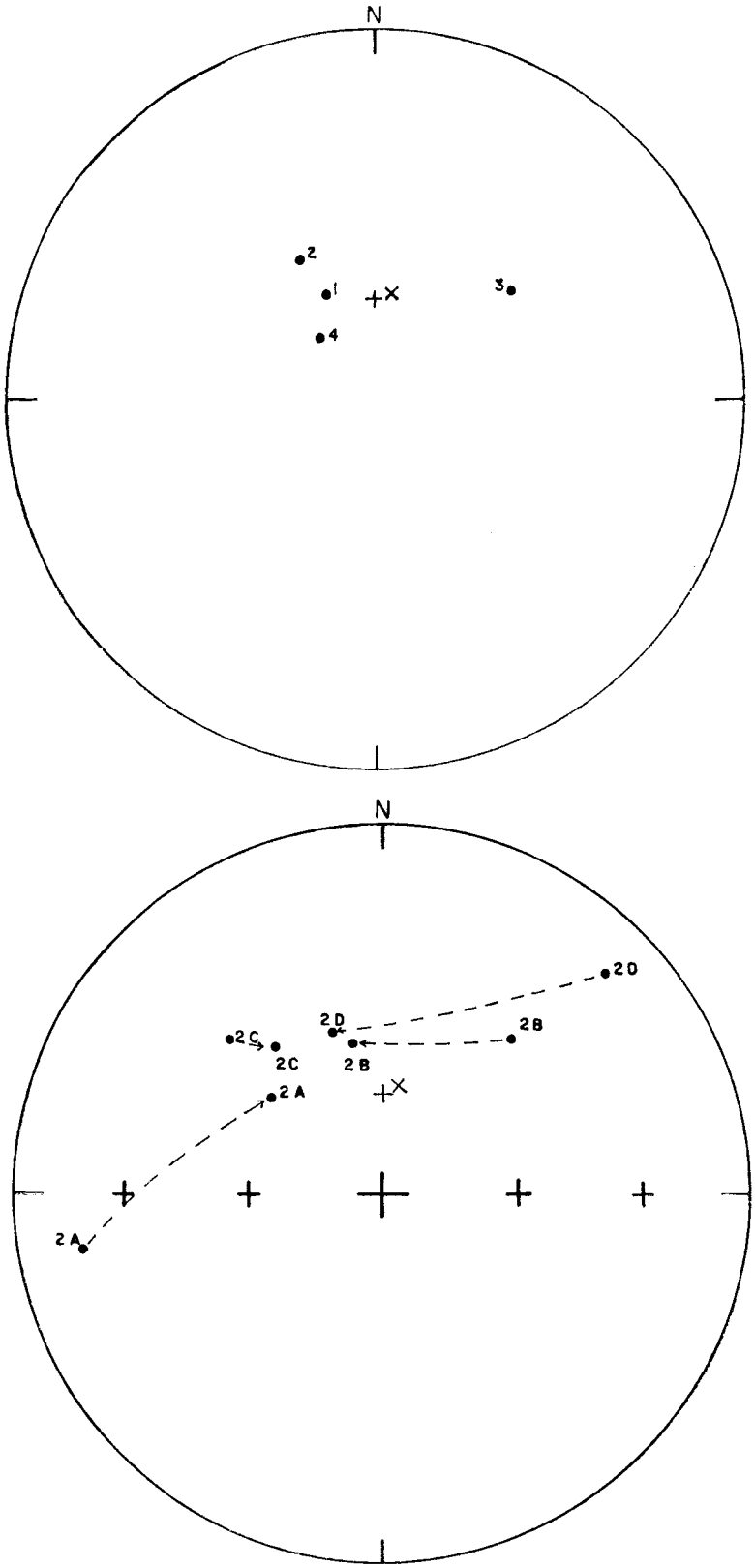


FIG. 2. (a) Palaeomagnetic directions of Tertiary intrusive (1) and associated baked Fountain rocks of Flagstaff Mountain (2, 3, 4). (b) Results of partial demagnetization of site No. 2 from baked zone.

Table 2

Directions of magnetizations, statistics and pole positions

Location	Ref.	Site	D_m	I_m	N	K	α_{95}	Polarity	Palaeopoles Lat.	Long.
HORN RANCH										
T4S,R83W	1	-100	156.2	-6.5	4	1693.2		R		
NE/4 SEC. 18	2	15	117.0	26.5	2	103.5		R		
39.7N,106.73W	3	85	157.3	-8.5	4	134.6		R		
	4	120	139.6	12.9	5	119.1		R		
	5	190	178.3	29.1	4	1138.1		R		
	6	255	146.4	31.5	5	194.3		R		
	7	340	343.3	28.4	6	56.7		N		
	8	430	339.7	24.5	5	69.9		N		
	*9	485	171.0	-14.4	3	124.3		R		
	*10	525	164.9	-11.2	6	109.4		R		
	11	596	348.2	34.3	4	19.6		N		
	12	597	354.3	38.8	5	164.3		N		
3 KIAMAN MEAN (1-6)			149.3	15.0	6	9.7	22.6	R	35N,	111E
3 POST-KIAMAN MEAN (7-12)			346.7	25.3	6	46.5	9.9	M	61N,	101E
SQUAW CREEK										
T5S,R83W	1	5	128.5	30.5	2	264.1		R		
S/2 SEC. 2	2	510	108.4	5.0	2	49.9		R		
39.6N,106.6W	3	610	137.0	-0.5	2	215.4		R		
	4	1110	118.0	13.0	2	1684.1		R		
	5	1417	141.5	2.0	2	772.6		R		
	6	1605	136.5	-8.5	2	74.0		R		
	7	1825	125.5	-6.0	2	527.5		R		
	8	1870	137.0	-5.0	2	1647.7		R		
	9	2000	124.1	-13.5	2	120.1		R		
	10	2045	115.0	10.7	3	5057.2		R		
	11	2150	140.5	-5.5	2	147.2		R		
	12	2180	149.0	1.5	2	773.0		R		
	13	2200	110.5	-5.0	2	1015.5		R		
	14	2225	133.0	9.0	2	1661.8		R		
	15	2235	139.5	-1.0	2	202.0		R		
	16	2335	135.5	-2.0	2	526.0		R		
	17	2600	161.5	14.0	2	2657.3		R		
	18	2662	165.3	2.7	3	25.2		R		
	19	2970	139.5	4.0	2	2628.6		R		
	20	3205	98.2	16.3	3	24.4		R		
	21	3345	128.3	32.4	2	17.3		R		
	22	3420	138.0	0.7	3	82.0		R		
	23	3765	352.2	33.5	3	76.2		N		
	24	3815	3.6	39.6	3	142.2		N		
	*25	4010	156.5	16.0	2	484.7		R		
	*26	4015	168.8	0.6	5	139.6		R		
	*27	4055	145.9	12.8	6	89.1		R		
2 KIAMAN MEAN (1-22)			132.4	4.4	22	17.2	7.7	R	30N,	131E
2 POST-KIAMAN MEAN (23-27)			344.7	7.6	5	8.4	27.9	M	52N,	99E

Table 2 (*continued*)

Location	Ref.	Site	D_m	I_m	N	K	α_{95}	Polarity	Palaeopoles Lat.	Long.
RED SANDSTONE CREEK										
T5S,R81W SEC. 1	*1	117	137.8	3.5	4	135.9		R		
39.6N,106.4W	2	785	6.9	-35.6	5	68.8		N		
	*3	809	152.5	16.5	2	229.4		R		
	4	880	9.7	-15.3	3	452.7		N		
	*5	1430	145.0	26.0	2	3282.6		R		
	*6	1560	148.8	-9.1	3	60.5		R		
	7	1643	336.9	28.4	5	10.0		N		
	*8	1707	126.3	12.2	4	47.2		R		
	*9	1905	147.2	2.4	5	15.8		R		
	10	2015	2.4	-32.7	4	36.9		N		
	11	2150	145.3	8.8	3	50.8		R		
	12	2283	137.4	0.2	5	190.6		R		
	13	2415	137.4	-10.7	8	126.9		R		
	14	2450	127.4	-6.5	2	155.5		R		
	15	2575	140.0	-3.5	2	526.7		R		
	16	2635	144.0	-4.5	2	6584.5		R		
	17	2750	136.0	1.5	2	292.1		R		
	18	2950	102.5	2.0	2	2626.7		R		
1 PRE-KIAMAN MEAN (1-10)			336.2	-11.6	10	8.3	17.9	M	39N,	105E
1 KIAMAN MEAN (11-18)			133.9	-1.6	8	29.6	10.4	R	33N,	133E
SIX MILE CANYON										
T1N,R71W										
N/2 SEC. 2	1	LK 755	180.1	-10.0	8	160.8		R		
40.1N,105.3W	2	LK 770	160.4	-17.7	6	71.5		R		
5 MEAN FOR SECTION-KIAMAN			170.4	-14.1	2	31.0	46.5	R	56N,	92E
PINE BROOK HILLS										
T1N,R71W	1	LK-5	151.0	-20.5	2	2911.8		R		
SE/4 SEC. 14	2	LK-1	116.8	-22.5	2	1146.4		R		
40.0N,105.3W	3	LK-2	128.5	-24.0	2	2716.0		R		
	4	LK-4	153.5	-13.0	2	13822.6		R		
4 MEAN FOR SECTION-KIAMAN			137.8	-20.7	4	21.9	20.1	R	43N,	101E
LEE HILL ROAD										
T1N,R71W	1	FT 4-8	159.7	-14.3	5	64.0		R		
E/2 SEC. 11	2	F TR 1	116.9	44.0	3	384.9		R		
40.1N,105.3W	3	F TR 2	233.9	-30.8	5	107.8		R		
	4	LK 1	207.3	-46.0	3	704.8		R		
	5	LK 2	246.3	-47.3	5	58.1		R		
	6	LK 3	164.6	-18.9	5	40.6		R		
	7	LK 5	161.1	-44.1	3	79.8		R		
	8	LK 6	155.1	1.8	5	34.4		R		
	9	LK 7	158.4	-18.0	3	180.2		R		
	10	LK 8	137.6	-11.7	3	61.0		R		
	11	LK 9-2	164.2	-20.9	4	60.3		R		
3 MEAN FOR SECTION-KIAMAN			168.8	-23.5	11	4.0	26.3	R	60N,	97E

Table 2 (*continued*)

Location	Ref.	Site	D_m	I_m	N	K	α_{95}	Polarity	Palaeopoles Lat.	Long.
DOWE PASS										
T3N,R70W	1	120	140.2	-3.2	3	13.1		R		
SECS. 21 + 28	2	187	143.6	-0.3	5	14.3		R		
40.3N,105.3W	3	250	127.1	15.4	5	167.3		R		
	4	294	140.5	11.6	4	81.0		R		
	5	345	128.2	0.7	3	137.3		R		
	6	477	156.8	-36.3	3	136.2		R		
	7	855	121.0	17.3	4	110.2		R		
	8	993	132.8	37.3	5	14.6		R		
2 MEAN FOR SECTION-KIAMAN			135.9	5.7	8	11.9	16.8	R	31N,	129E
FLAGSTAFF MOUNTAIN										
T1N,R71W										
SW/4 SEC. 36	1	25	139.8	5.5	4	54.3		R		
40.0N,105.3W	2	99	156.5	-17.6	5	44.6		R		
1 MEAN FOR SECTION-KIAMAN			148.0	-6.1	2	16.4	66.4	R	43N,	121E
A KIAMAN MEAN-WESTERN AREA			138.4	6.0	3	42.3	19.2	R	33N,	126E
B KIAMAN MEAN-EASTERN AREA			151.9	-12.1	5	16.7	19.3	R	48N,	119E
C POST-KIAMAN-WESTERN AREA			345.7	16.5	2	41.5	39.8	M	56N,	100E
FLAGSTAFF MTN.										
1 TERTIARY INTRUSIVE			336.2	64.8	6	194.7	4.9	N	72N,	166W
2 BAKED ZONE			332.9	54.7	4	59.2	12.1	N		
3 BAKED ZONE (LIGHTNING?)			51.4	50.5	5	3.6	43.6	N		
4 1 FT. ABV. BAKED ZONE			319.9	72.0	5	137.4	6.6	N		

* Directions rotated 180° to calculate the mean.

60 to 70 million years ago. Since this time the magnetic field has had a high inclination, so that the rocks examined have been subjected for a prolonged period of time to a magnetic field with a different orientation from the direction of magnetization shown by them. When the structure is corrected back to its probable original orientation, consistent field directions are found, indicating that magnetization is pre-folding.

Numerous samples were subjected to a.c. demagnetization. The samples chosen for demagnetization were generally those that showed considerable scatter or those whose directions deviated significantly from the major grouping. The latter belong to two classes: (1) those with directions which were well grouped within a site but unrelated either to the major grouping or to the present-day field direction; (2) those with directions essentially aligned with the present-day field. The latter class invariably become unmeasurable upon demagnetization in fields of 50 Oe. During a storage test, a specimen of this type became magnetized in the ambient field which was approximately 180° opposed to the original direction measured in the sample. Samples chosen for demagnetization were demagnetized in steps of 50 Oe. Most samples of this type appear to have a soft magnetization which can be removed in fields of 250 Oe. The response was very similar to that illustrated in the baked zone of the Fountain (Fig. 2(b)). A few of these samples appear to have a harder magnetization and did not respond to a.c. demagnetization in fields of 300 Oe (3, 4 and 5 of Fig. 5(c)).

Comparison of Pre-Kiaman rocks of mixed polarity from the Red Sandstone Creek section shows a mean difference of 144° between normal and reversed rocks.

A similar comparison of Post-Kiaman rocks from Horn Ranch and Squaw Creek show a mean difference of 161° and 130°, respectively, between normal and reversed rocks. If we assume that reversals of the Earth's field are about 180° (Cox & Doell 1960) the discrepancies may be due in part to the addition of a viscous magnetization component, and possibly partly to inadequate structural correction. Inadequate sampling could also explain directions that are not exactly 180° apart. It has been suggested (Irving 1966) that the field may not reverse exactly, so that the directions reported here may be a reflection of the field at the time. In any event, the directions indicate some degree of stability since the time at which the normal and reversed magnetizations were acquired.

A comparison of the intensities of NRM of the rocks showing both normal and reversed polarities is made in Fig. 3. Fifty-seven specimens showing negative polarity and 182 specimens showing positive polarity were used. The two histograms are similar in that the greatest percentage of samples of each group have intensities between $J = 1\times10^{-6}$, and $J = 2\times10^{-6}$ gauss. In each group approximately 17 per cent of the samples have intensities below 1×10^{-6}. The absence of intensities below the level of $0{\cdot}2\times10^{-6}$ gauss reflects an inability to measure lower intensities.

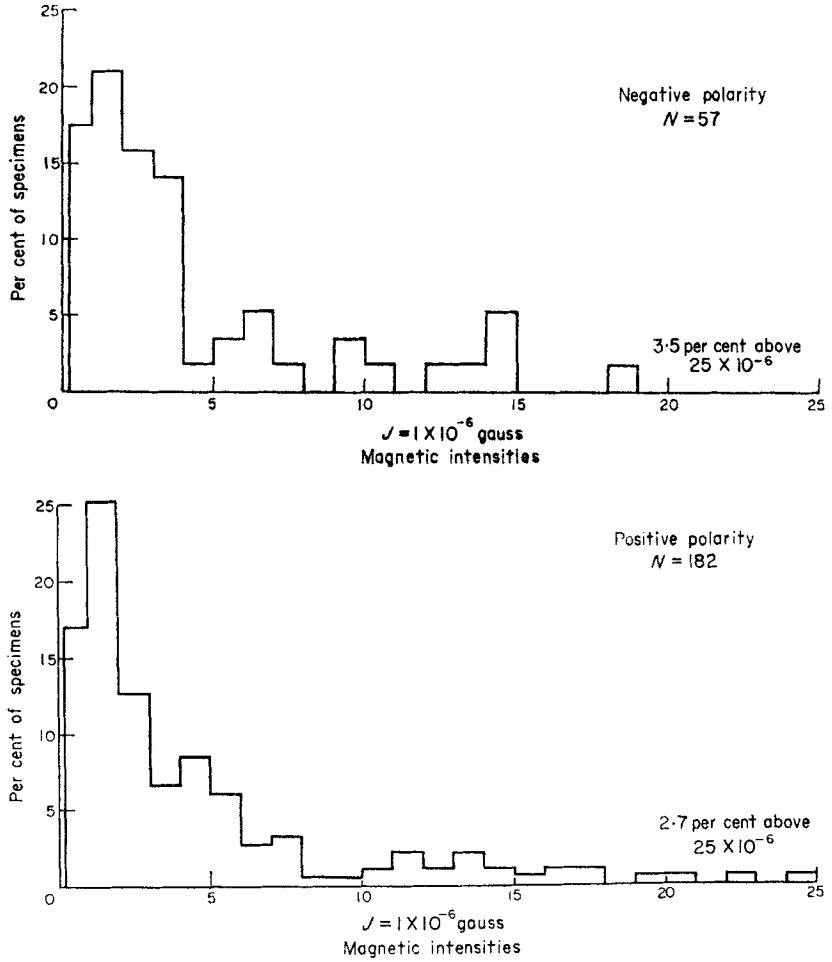


FIG. 3(a). Distribution of intensities for specimens with normal polarity. (b) Distribution of intensities for specimens with reversed polarity. N is the total number of samples considered in each group.

Eighty-three per cent of the samples with positive polarity and 81 per cent of the samples with negative polarities have intensities less than 8×10^{-6} gauss. The similarity of intensity distribution of reversed and normal samples indicates no consistent difference between the two suites of rocks.

Because of these tests it appears that the magnetization has been stable for a long period of time, but it is not possible to know whether this magnetization was acquired at the time of deposition.

Magnetic data

Each of the equal area projection nets shows the mean direction for each site sampled and the mean for the section (Figs 4 and 5). Two to eight specimens were examined from each site. Details are in Table 2. For the three western sections two mean directions are given. The mean for the Pre- and Post-Kiaman zones of mixed polarity are shown in the northern hemisphere. It is important to note that the Pre- and Post-Kiaman sites having reversed magnetic directions are plotted in the southern hemisphere, although in calculating the mean, the directions have been rotated 180° and averaged with the normal directions.

The individual directions shown in Fig. 6(a) are the mean directions obtained from the western area. The mean shown in the southern hemisphere is the mean of the Kiaman magnetic directions from the western area. Fig. 6(b) shows the mean magnetic directions for the Kiaman of the eastern area and Fig. 6(c) compares the Kiaman magnetic directions of the eastern and western areas.

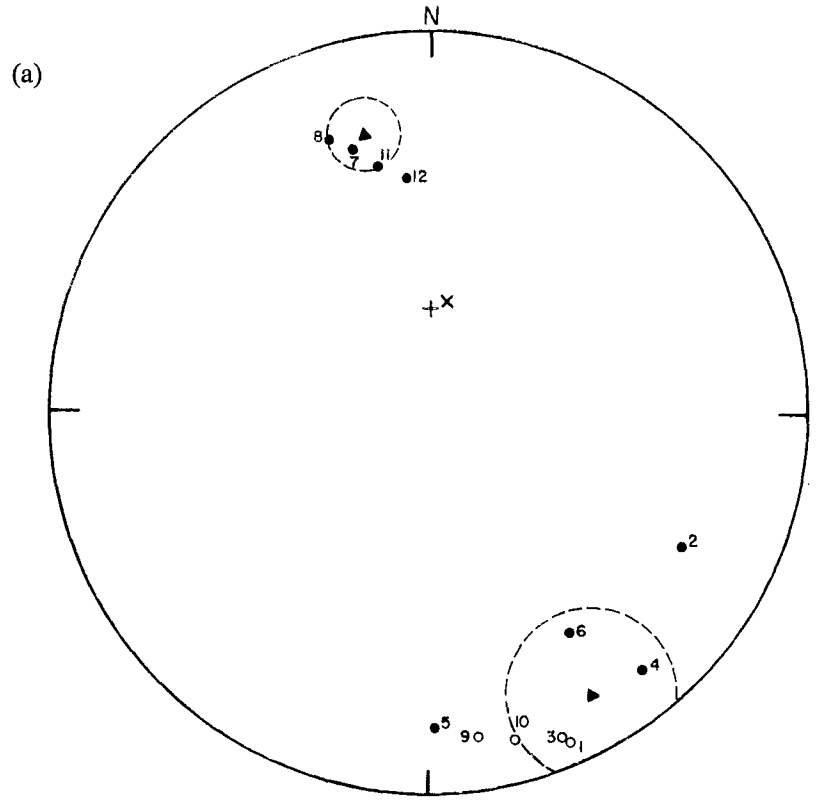


FIG. 4(a). Magnetic data—Horn Ranch—Maroon formation.

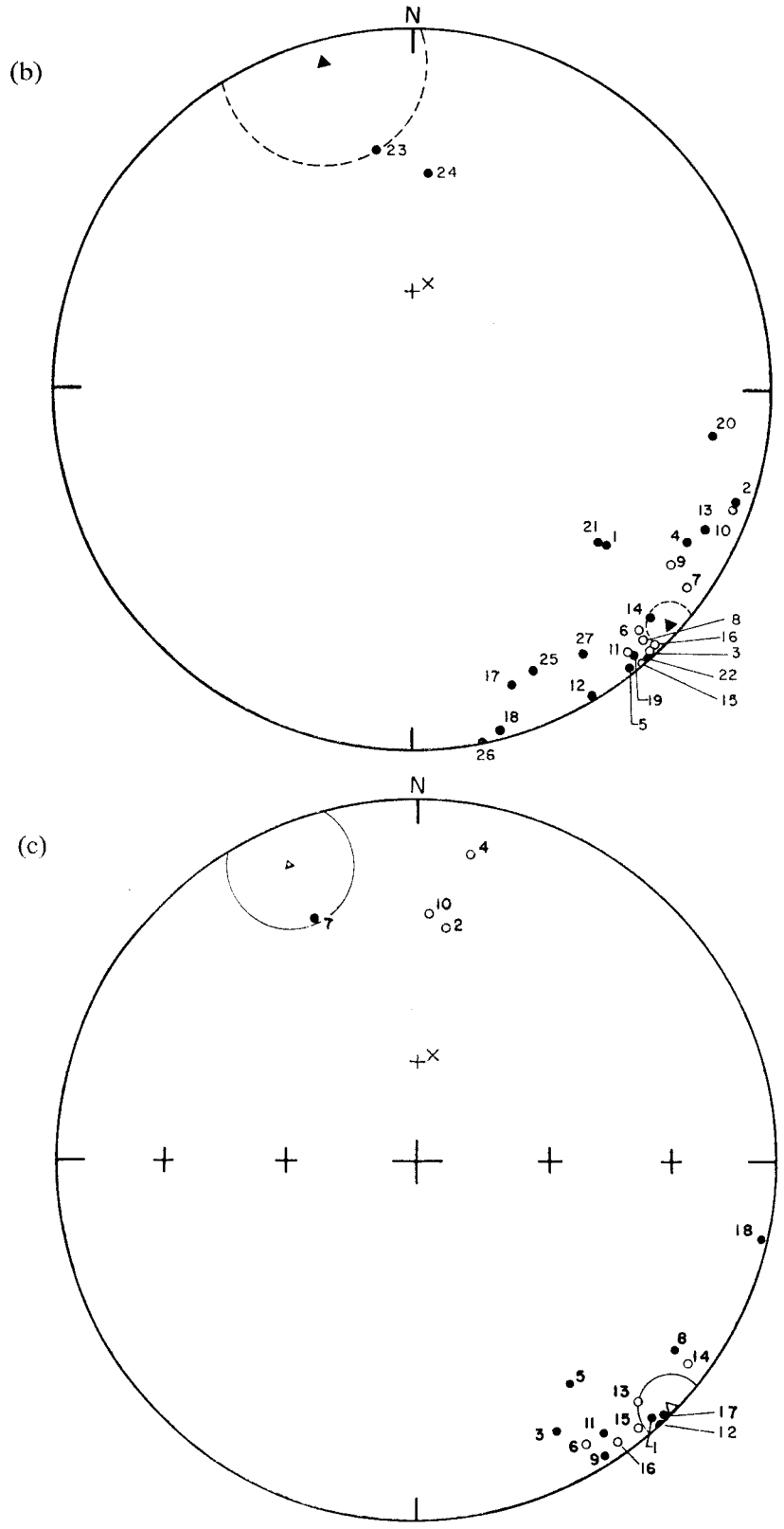


FIG. 4(b). Magnetic data—Squaw Creek—Maroon formation. (c) Magnetic data—Red Sandstone Creek—Minturn and Maroon formations.

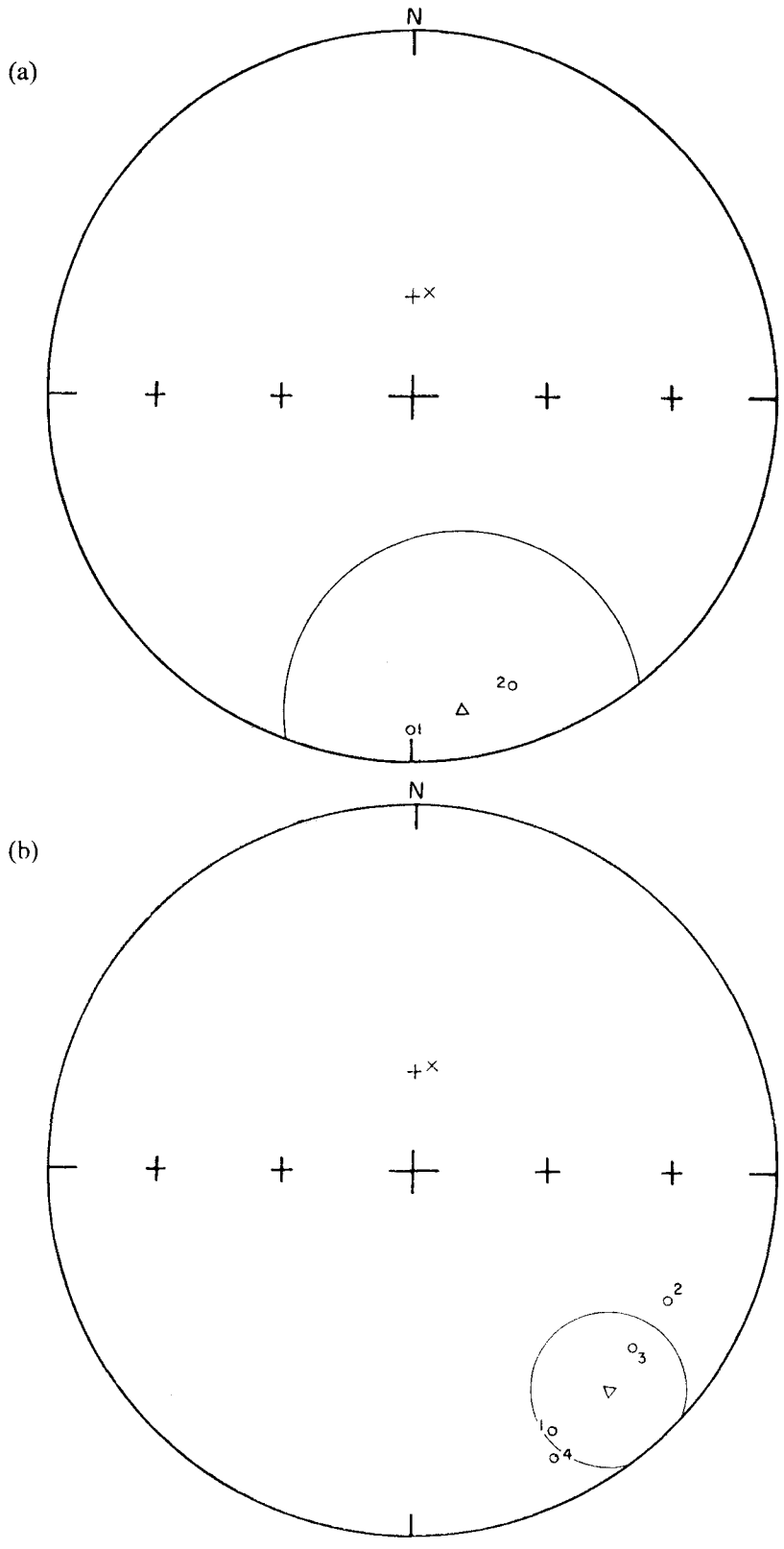


FIG. 5(a). Magnetic data—Six Mile Canyon—Lykins formation. (b) Magnetic data—Pine Brook Hills—Lykins formation.

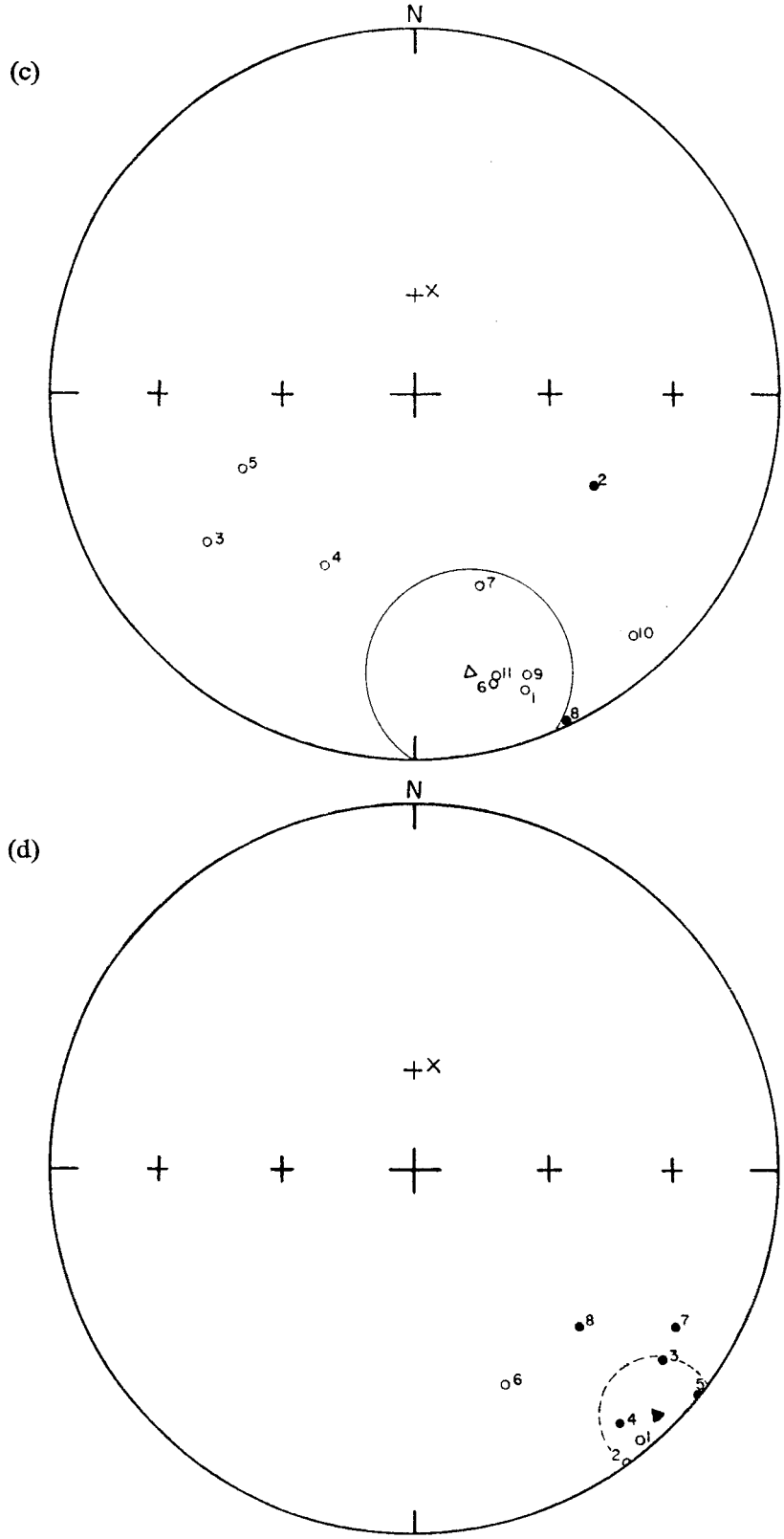


FIG. 5(c). Magnetic data—Lee Hill Road—Fountain and Lykins formations.
(d) Magnetic data—Dowe Pass—Fountain formation.

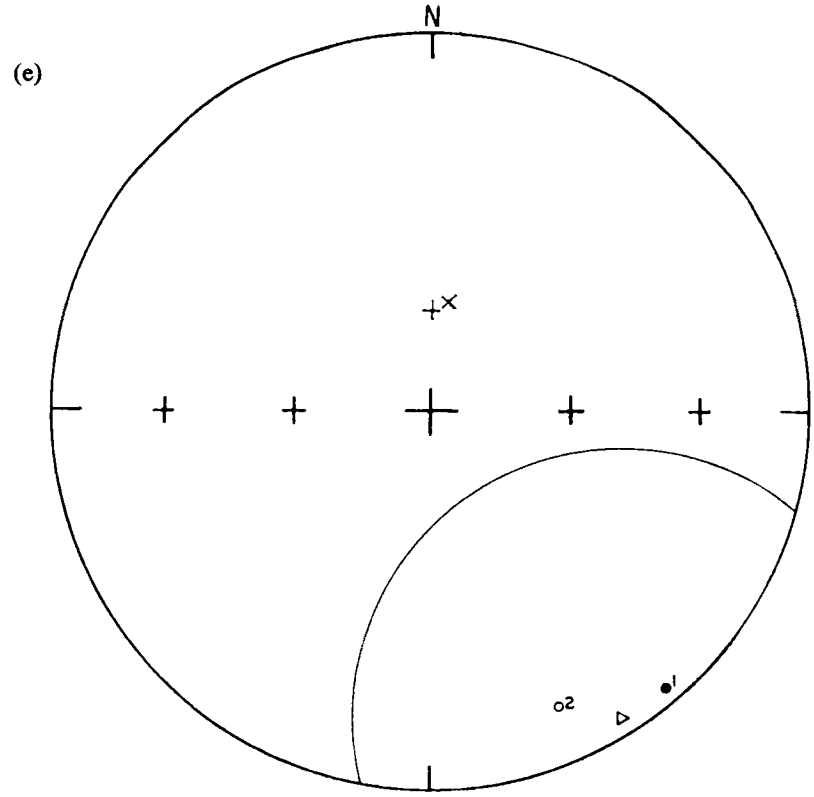


FIG. 5(e). Magnetic data—Flagstaff Mountain—Fountain formation.

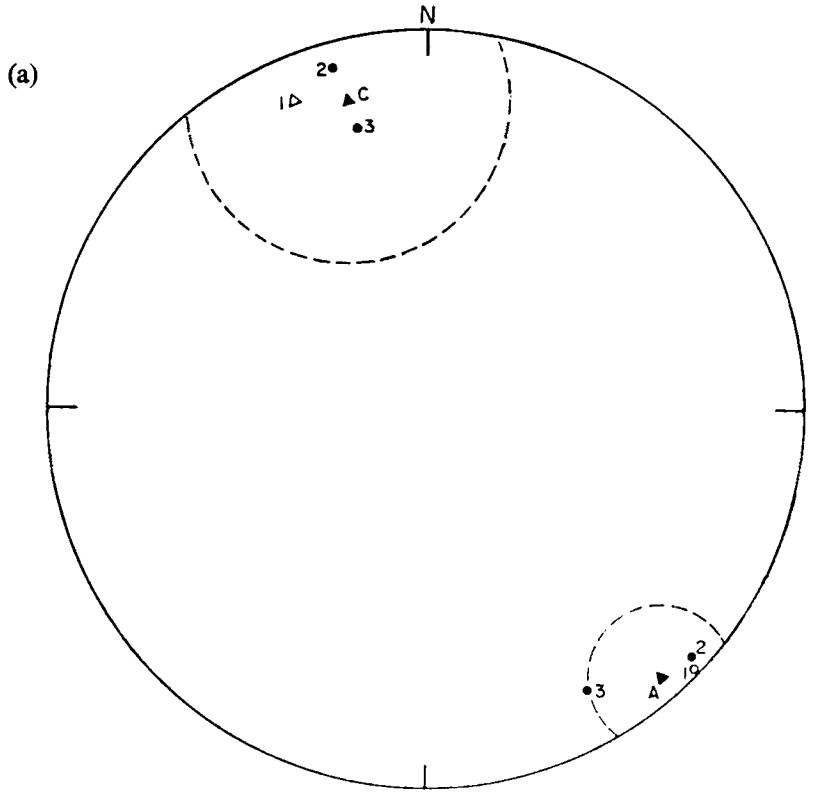


FIG. 6(a). Mean directions from western area.
Δ Pre-Kiaman—Red Sandstone Creek.
A: Kiaman mean
1. Red Sandstone Creek.
2. Squaw Creek.
3. Horn Ranch.
C: Post-Kiaman mean
2. Squaw Creek.
3. Horn Ranch.

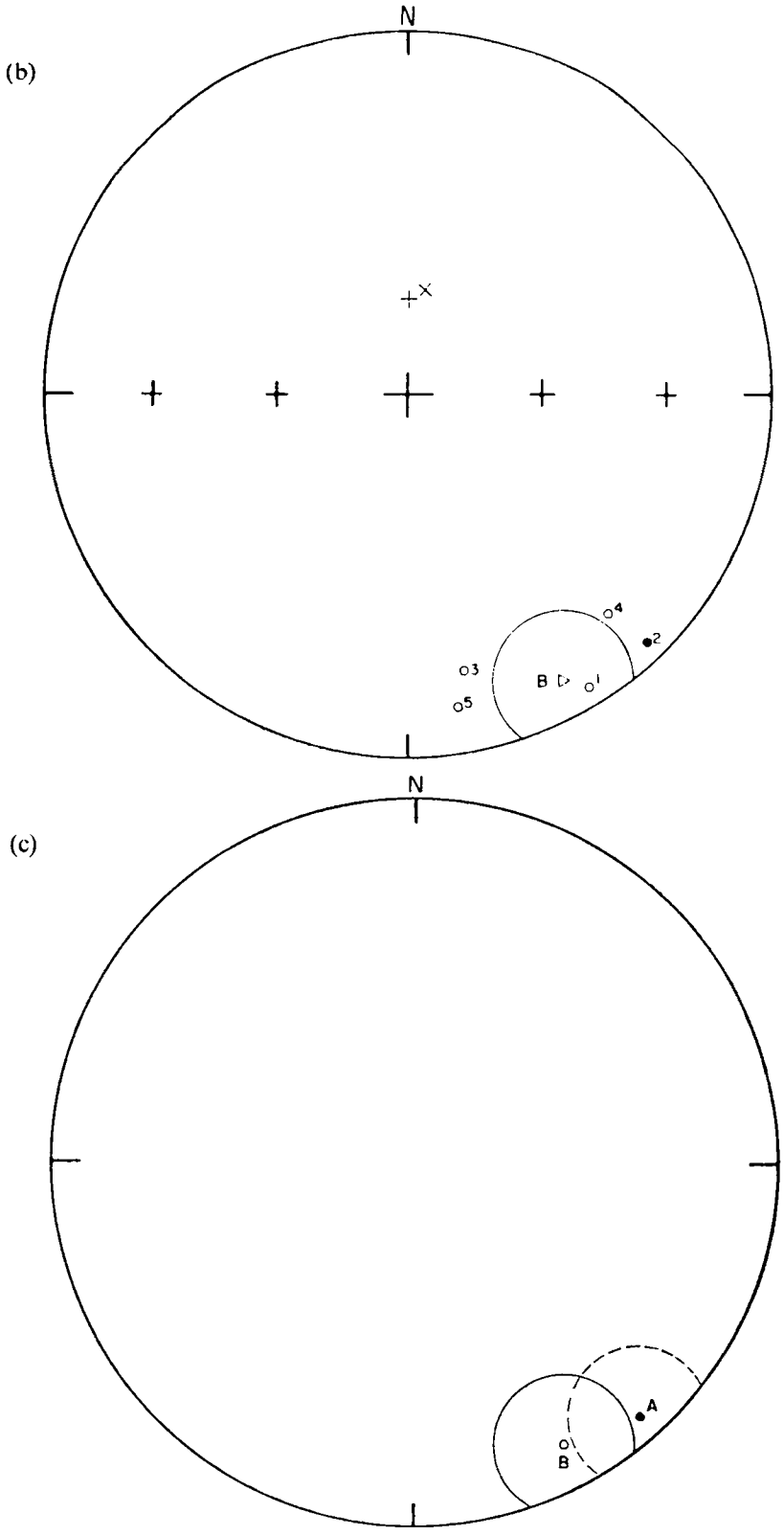


FIG. 6(b). Mean directions from eastern area.
B: Kiaman
1. Flagstaff Mountain.
2. Dowe Pass.
3. Lee Hill Road.
4. Pine Brook Hill.
5. Six Mile Canyon.

(c) Overall Kiaman means.
A: western area.
B: eastern area.

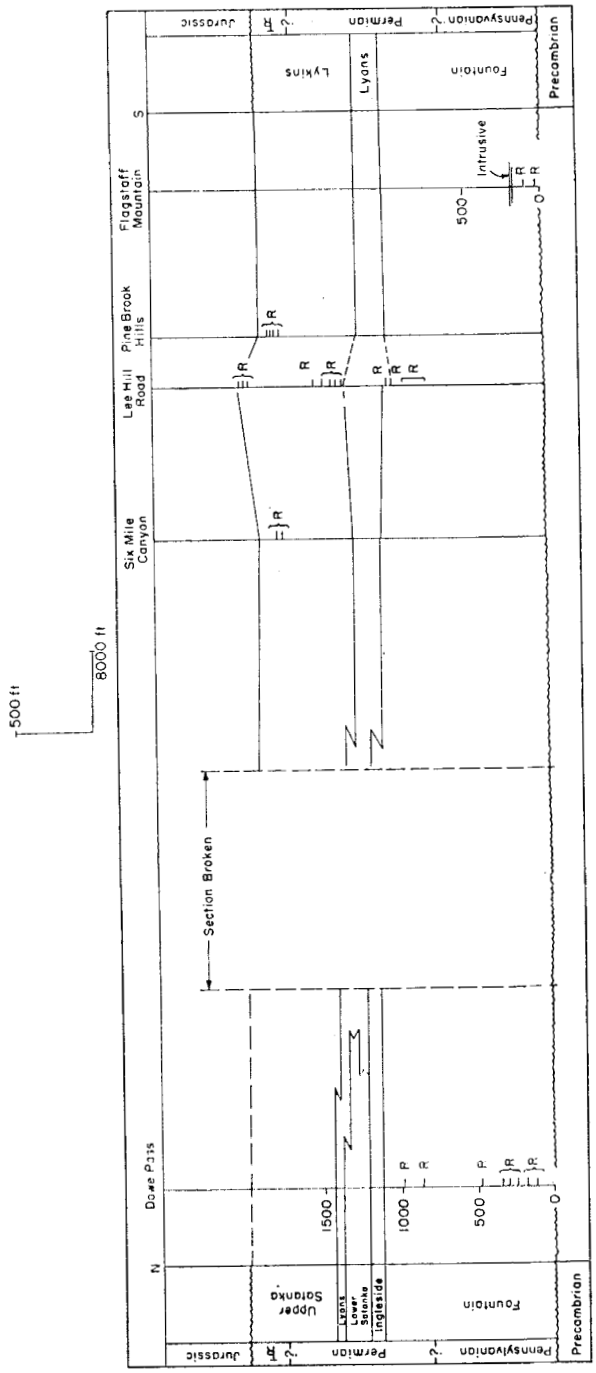


FIG. 7. Cross-section of Boulder area.

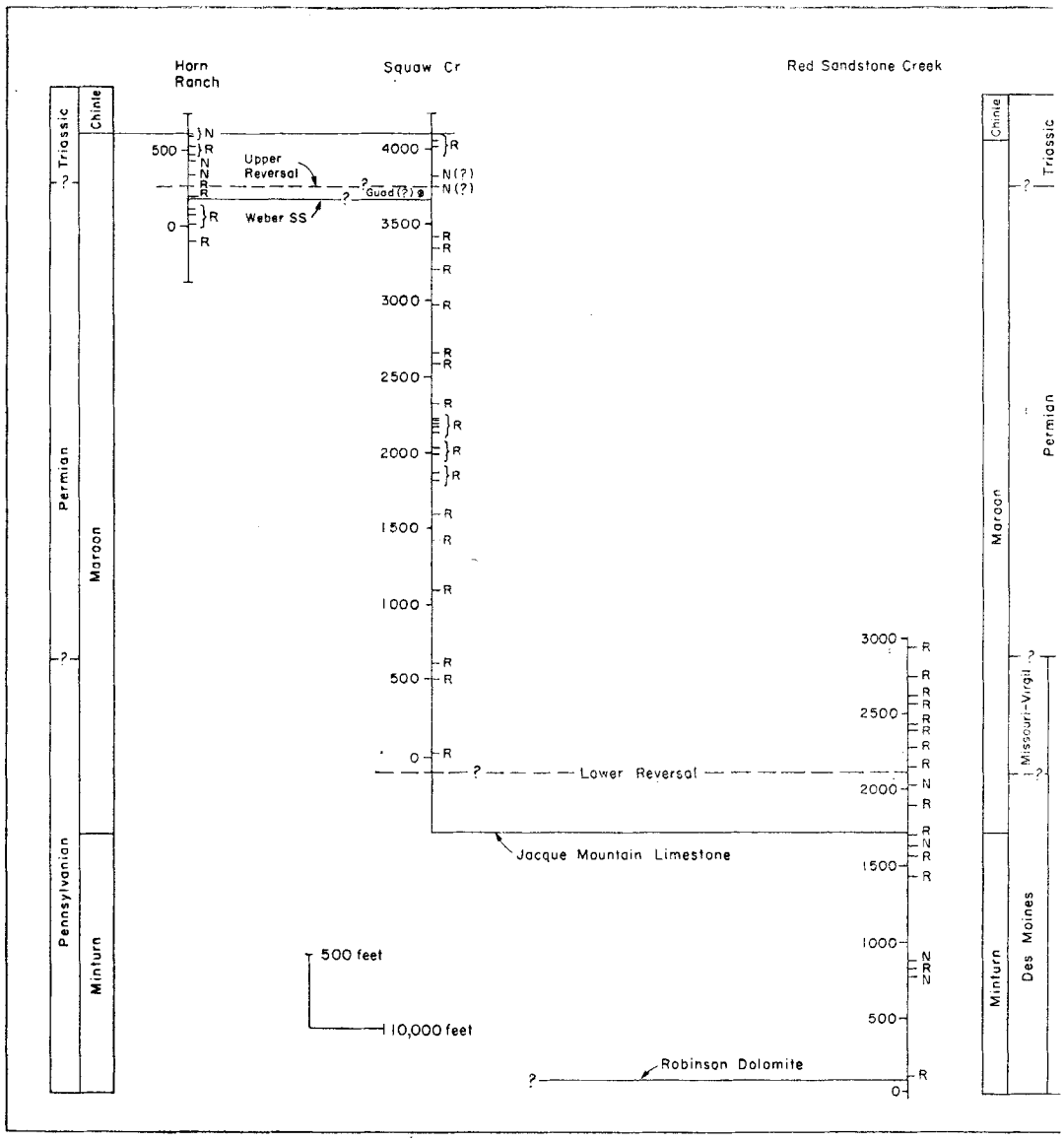


FIG. 8. Cross-section of Eagle area.

Discussion

The stratigraphic relationships of the collection sites and the approximate level of the reversals are shown in Figs 7 and 8. It is also indicated on the figure whether a particular site has normal (*N*) or reversed (*R*) directions. In the stratigraphic intervals identified as Pre-Kiaman and Post-Kiaman, the normal and reversed directions occur with about the same frequency (Fig. 8). By contrast there is a complete absence of reversals through a long stratigraphic section on both the eastern and western slopes, and this observation supports the interpretation that the Kiaman division has been located. The stratigraphic distance between sampling sites in this study is too great to be sure that no reversals occur in this zone. It is of interest to note that Helsley (1965) has located some normally magnetized rocks near the Penn-

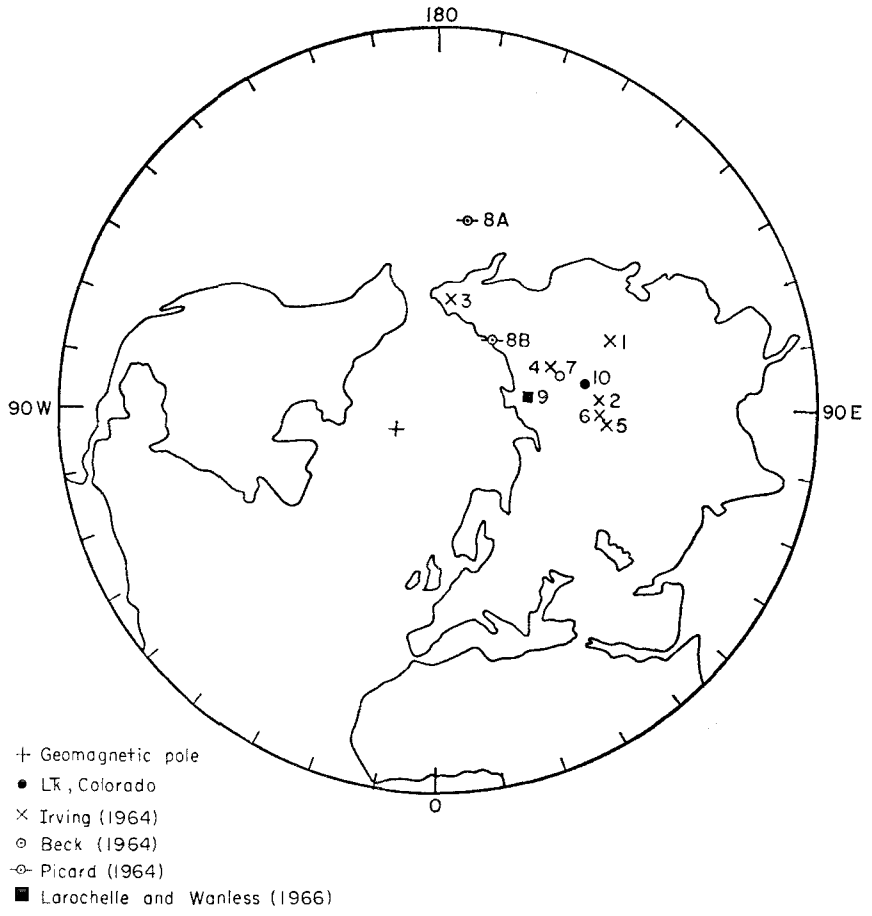


FIG. 9 (a). Palaeomagnetic poles for North America—Triassic period.

sylvanian–Permian boundary in the Dunkard Series of West Virginia, and Graham (1955) reported two normally magnetized specimens from a collection of the Supai formation obtained at Oak Creek Canyon, Arizona (Fig. 1). As the lower part of the Supai at this location may include Upper Pennsylvanian and Lower Permian units, (Jackson 1951) rocks with normal magnetization from this location may be the same age as those from the Dunkard Series. Although it is tempting to interpret these normal directions in rocks believed to be Permian age as a short-lived normal epoch during the Kiaman interval, they may be examples of remagnetization or be due to stratigraphic miscorrelation.

If this long reversal found in the redbeds represents the Kiaman interval, the stratigraphic relationships (Figs 7 and 8) indicate that the Kiaman began near the beginning of the Missourian Epoch (Late Pennsylvanian) and continued until Middle Scythian (Early Triassic) time (McMahon & Strangway 1967). This time span is in good agreement with that postulated by Irving & Parry (1963) who suggested that the Kiaman extended from the beginning of the Stephanian Stage until the approximate close of the Permian Period. Scattered collections made from other formational units in North America of the appropriate age (Fig. 1) have yielded only reversed magnetic directions with the exceptions in the Supai and Dunkard noted above.

The Russians report mixed directions from the Upper Tartarian beds of the Late Permian age (Petrova 1965) and from a Lower Permian greenish-gray sandstone of the Abakan trough (Popova 1963). To date, sequences have not been located and examined which confirm the presence of these zones in North America and Europe. More palaeomagnetic data on this time span is necessary before the Russian data can be fully integrated with the data from North America.

All of the units examined palaeomagnetically on the North American continent consist of red sedimentary rocks. These beds have been dated either on the basis of their own fossil assemblages or by their stratigraphic relationship to formations which have been dated on palaeontological criteria. The units in Australia, upon which the definition of the Kiaman interval is principally based, are igneous rocks which have been dated as Permian both by radioisotope dating methods and by their stratigraphic position in the Australian sedimentary sequence. The excellent age agreement of the North American sedimentary rocks which appear to belong to the Kiaman division and the Australian igneous rocks of the type locality, which owe their magnetic moments to thermoremanent magnetization suggests that the equivalent redbeds of North America received their magnetization early in their history. Their magnetic direction may be due to a detrital remanent magnetization or chemical alteration shortly after deposition.

Palaeopole positions calculated from the present data are in good agreement with the data for the Permian and Triassic of North America presented by Irving

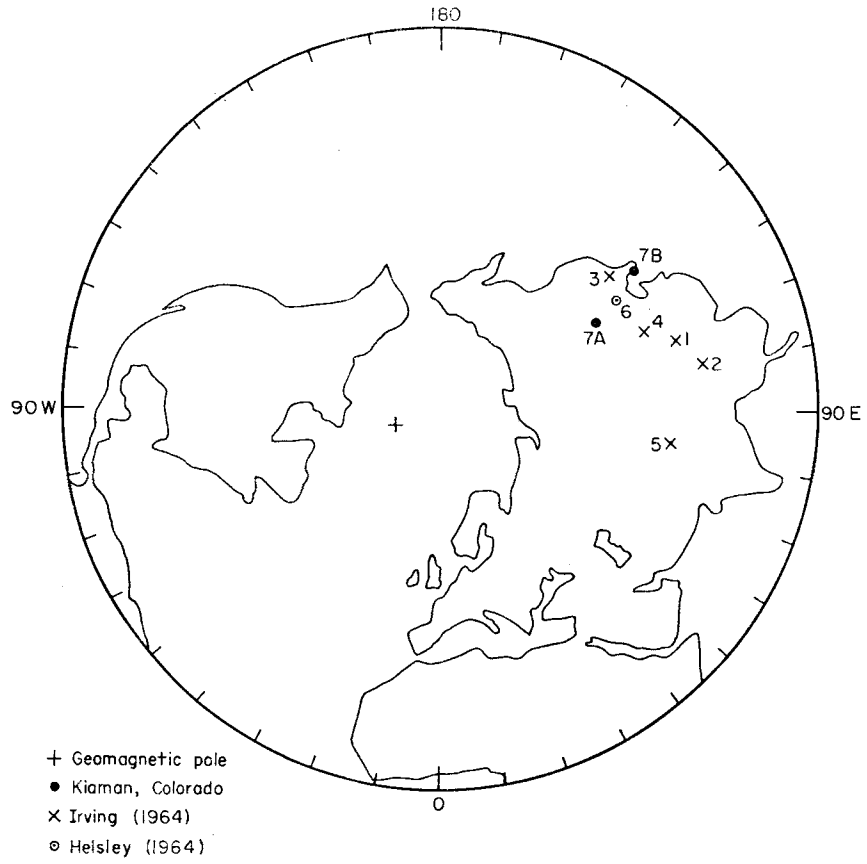


FIG. 9(b). Palaeomagnetic poles for North America—Permian period.

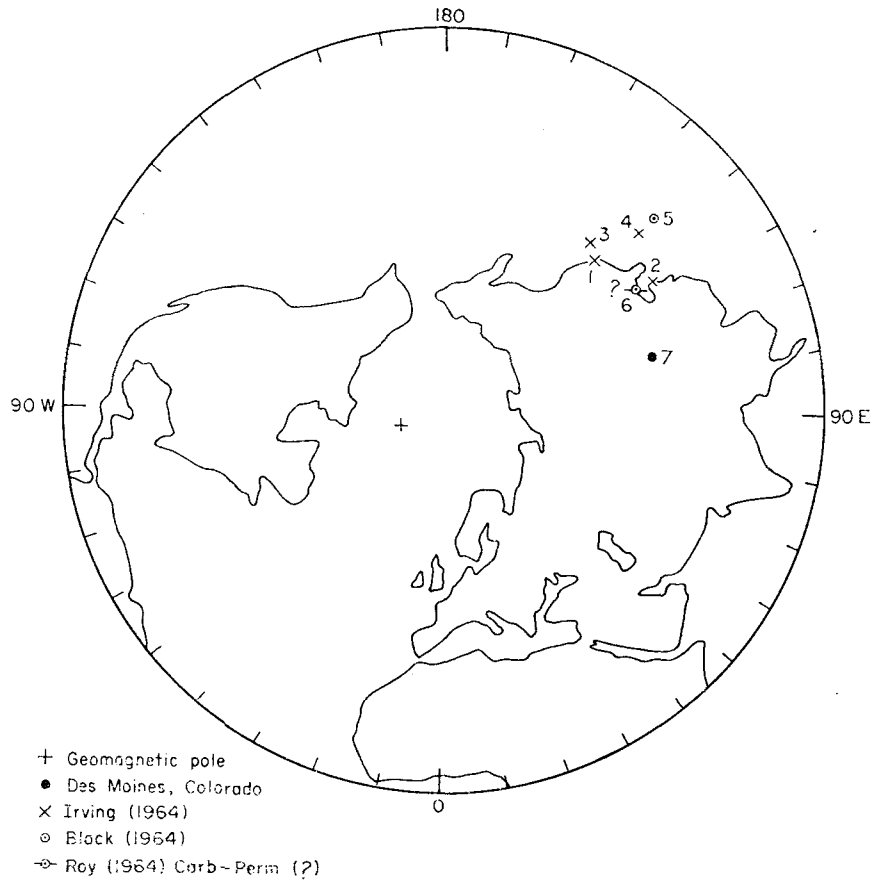


FIG. 9(c). Palaeomagnetic poles for North America—Mississippian and Pennsylvanian periods.

(1964) (see Table 3). The palaeomagnetic pole for the Early Triassic of Colorado is 56° N, 100° E. Irving (1964) calculates a mean position of the Triassic palaeomagnetic pole for the North American continent at 59° N, 106° E. Fig. 9(a) reproduces Irving's B group data for the North American continent (Irving 1964, p. 118). To these data have been added the Lower Triassic palaeopole calculated from the present data, palaeopoles for the Chugwater of Wyoming (Picard 1964), the palaeopole of a Triassic diabase of southwestern Pennsylvania (Beck 1965) and the palaeopole given by Larochelle & Wanless (1966) for a Triassic dike in Nova Scotia.

The palaeomagnetic pole for the Kiaman interval based on data from the eastern slope is located at 48° N, 119° E, and on the basis of data from the western slope is 33° N, and 126° E. Irving (1964) calculates a mean position of the Permian magnetic pole for the North American continent to be 38° N, 105° E. Fig. 9(a) reproduces Irving's B group data (Irving 1964, p. 119). To these data have been added the palaeopoles calculated from the eastern and western slopes of the Colorado Front Range and the palaeopole calculated for the Dunkard (Helsley 1965).

The palaeomagnetic pole for the Pre-Kiaman (Des Moinesian Epoch of the Pennsylvanian) is calculated to be located at 39° N, 105° E. Fig. 9(c) reproduces Irving's B group data for the Carboniferous (Irving 1964, p. 120). To these data have been added the pole positions calcuated for the Pre-Pictou sands of Newfoundland (Black 1964), the Red Beds of Prince Edward Island which are dated as Carbon-

Table 3

Pole positions for North American Triassic Permian and Carboniferous formations

Ref. No.	Locality sampled	Palaeopoles Lat.	Long.	References
Triassic				
1	Chugwater fm; western U.S.	48°N	112°E	*Irving 8:30
2	Chinle fm; western U.S.	55 N	93 E	*Irving 8:40
3	New Oxford; eastern U.S.	66 N	174 E	*Irving 8:41
4	Newark; eastern U.S.	63 N	108 E	*Irving 8:42
5	Conn. Valley; eastern U.S.	54 N	86 E	*Irving 8:43
6	Mass. lava; eastern U.S.	55 N	88 E	*Irving 8:44
7	Diabase, Pa.	62 N	105 E	Beck (1964)
8a	Chugwater fm; Wy. intervals; 7, 5, 4, 3, 1	48 N	178 E	Picard (1964)
8b	Chugwater fm; interval; 6	70 N	142·5 E	Picard (1964)
9	Nova Scotia dike	69 N	98 E	Larochelle & Wanless (1966)
10	Upper Maroon (equiv. Moenkopi)	56 N	100 E	McMahon & Strangway (1967)
Permian				
1	Cutler fm; southwest U.S.	34 N	107 E	*Irving 7:46
2	Abo fm; southwest U.S.	30 N	100 E	*Irving 7:49
3	Yeso fm; southwest U.S.	41 N	127 E	*Irving 7:50
4	Supai fm; southwest U.S.	40 N	110 E	*Irving 7:51
5	Sangre de Cristo fm; Colorado	38 N	81 E	*Irving 7:52
6	Dunkard, West Virginia	44 N	122·3 E	Helsley (1965)
7a	Fountain and Lykins fms; Colorado	48 N	119 E	McMahon & Strangway (1967)
7b	Maroon fm; Colorado	33 N	126 E	McMahon & Strangway (1967)
Carboniferous				
1	Barnett fm; Texas	41 N	135 E	*Irving 6:63
2	Naco fm; southwest U.S.	35 N	122 E	*Irving 6:66
3	Codroy fm; Newfoundland	37 N	139 E	*Irving 6:67
4	Eastern Canada	30 N	133 E	*Irving 6:68
5	Pre-Pictou Ss; Newfoundland	24 N	133 E	Black (1964)
6	Redbeds; Prince Edward Island (Carb.–Perm.)	39 N	123 E	Roy (1963)
7	Upper Minturn fm; Colorado (Des Moines)	39 N	105 E	McMahon & Strangway (1967)

* Irving (1964).

iferous to Permian (Roy 1963) and the pole position calculated for the Des Moinesian sequence at Red Sandstone Creek. It should be noted that most of the pole positions shown in Fig. 9(c) are for units of Mississippian age and that the pole from Colorado is based on data recovered from the upper 1700 ft of several thousand feet of section, all of which is assigned to the Des Moinesian Epoch.

Conclusions

The Kiaman magnetic division has now been spanned in two sections located on two continents. It is possible that the Colorado section contains a more complete rock sequence of the zone than is present in Australia (Irving 1966).

The present study essentially supports the original interpretation concerning the

age and duration of the interval and the stability of the polar pattern during this interval.

The approximate stratigraphic levels of the top and bottom of the division have been located in the Colorado section. With the possible exception of Picard's work on the Chugwater in Wyoming (1964), this study represents the only instance in which either limit of the Kiaman has been crossed on the North American continent.

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References

Beck, Jr, M. E., 1965. Palaeomagnetic and geologic implications of measured magnetic properties of the Triassic diabase dike of southeastern Pennsylvanian, *J. geophys. Res.*, **70,** 2845–2856.

Black, R. F., 1964. Palaeomagnetic support of the theory of rotation of the western part of the Island of Newfoundland, *Nature, Lond.*, **202,** 945–948.

Collinson, D. W., 1965. The remanent magnetization and magnetic properties of red sediments, *Geophys. J. R. astr. Soc.*, **10,** 105–126.

Cox, A. & Doell, R. R., 1960. Review of palaeomagnetism, *Bull. geol. Soc. Am.*, **71,** 645–768.

Graham, J. W., 1955. Evidence of polar shift since Triassic time, *J. geophys. Res.*, **60,** 329–347.

Helsley, C. E., 1965. Palaeomagnetic results from the Lower Permian Dunkard Series of West Virginia, *J. geophys. Res.* **70,** 413–424.

Hide, R., 1966. Free hydromagnetic oscillations of the Earth's core and the theory of the geomagnetic secular variation, *Phil. Trans. R. Soc.*, **A259,** 615–650.

Irving, E., 1964. *Palaeomagnetism and its applications to Geological and Geophysical Problems*, p. 399, John Wiley, New York.

Irving, E., 1966. Palaeomagnetism of some Carboniferous rocks from New South Wales and its relation to geologic events, *J. geophys. Res.*, **71,** 6025–6051.

Irving, E. & Parry, L. G., 1963. The magnetism of some Permian rocks from New South Wales, *Geophys. J. R. astr. Soc.*, **7,** 385–411.

Jackson, R. L., 1951. Stratigraphic relationships of the Supai formation of central Arizona, *Plateau*, **24,** 84–91.

Larochelle, A. & Wanless, R. K., 1966. The palaeomagnetism of a Triassic diabase dike in Nova Scotia, *J. geophys. Res.*, **71,** 4949–4953.

McMahon, B. E. & Strangway, D. W., 1968. *Bull. geolog. Soc. Am.*, in press.

Petrova, G. N., 1965. On validity of palaeomagnetic data and transitional regime of geomagnetic field, *J. Geomagn. Geoelect. Kyoto*, **17,** 425–434.

Picard, M. A., 1964. Palaeomagnetic correlation of units within Chugwater (Triassic) formation, west-central Wyoming, *Bull. Am. Ass. Petrol. Geol.*, **48,** 269–291.

Popova, A. V., 1963. Palaeomagnetic investigations of Palaeozoic sedimentary rocks in Siberia, *Akad. Nauk. SSSR Bull. Geophys. Ser.*, p. 274–277.

Roy, J. L., 1963. Palaeomagnetism of Prince Edward Island, *Geophys. J. R. astr. Soc.*, **8,** 226–229.